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ML-Assisted Nusselt Correlations for Flow Boiling of FC-72 and FC-770 in Rectangular Minichannels

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This study presents an experimental and data-driven procedure for deriving local Nusselt number correlations for flow boiling in rectangular minichannels. The analysis is demonstrated primarily for FC-72 and supported by a smaller FC-770 experimental series processed using the same protocol. Local heat-transfer data were obtained from infrared wall-temperature measurements combined with pressure, temperature, flow rate and electrical power measurements. The data reduction included heat-loss correction, wall-conduction correction, local saturation-temperature reconstruction and regime classification into subcooled and saturated regimes. Tree-ensemble models were used as an auxiliary tool to identify nonlinear trends and support the selection of compact explicit correlation forms. For the FC-72 series, approximately 200,000 quality-controlled points per regime were used for tree-ensemble model development and validation. Extra Trees achieved mean MAPE values of 0.170 ± 0.025 for the subcooled regime and 0.174 ± 0.050 for the saturated regime under experiment-wise validation. For the larger FC-72 dataset, regime-specific power-law correlations were then fitted using dimensionless groups and axial position. The saturated-boiling correlation showed the most stable performance, whereas the subcooled-boiling correlation should be treated as an empirical approximation within the investigated operating range. The proposed approach provides a consistent basis for later extension to additional fluids and surface conditions.

Introduction

Flow boiling in mini- and microchannel heat exchangers is a promising route to compact thermal management when high heat fluxes must be removed within a limited footprint (e.g., electronics cooling and compact power systems). In such confined geometries, heat transfer is tightly coupled to evolving two-phase flow structures; therefore, local information is required to interpret thermal performance, compare operating conditions, and develop reliable predictive models.

A key performance metric in convective heat-transfer equipment is the heat transfer coefficient (HTC), as it quantifies the intensity of wall–fluid heat transfer and directly informs heat exchanger design and assessment. In electrically heated test sections, the local HTC is commonly defined at the heated wall as the ratio of local heat flux to an appropriate temperature driving force, providing a spatially resolved measure of the wall–fluid thermal interaction. The HTC also underpins the Nusselt number, which enables dimensionless comparison across fluids, geometries, and operating conditions. Accordingly, a large body of theoretical and semi-empirical correlations has been developed to predict Nu (and thus HTC) in terms of similarity parameters such as Reynolds and Prandtl numbers and, for boiling flows, additional groups associated with phase change and interfacial dynamics. In practice, however, predictive accuracy often degrades when correlations are applied outside their original ranges of fluids, channel sizes, orientations, heating configurations, or flow regimes. This motivates careful experimental determination of local HTC under well-defined conditions and transparent reporting of applicability limits.

Machine-learning (ML) methods are increasingly used in mini-/microchannel heat-transfer research because the governing relationships are strongly nonlinear, regime-dependent, and often geometry-specific, which limits the portability of classical correlations across datasets and test sections. Recent reviews frame ML as (i) a data-driven modelling layer for mini-/microchannel problems (single-phase and two-phase), including sensitivity analysis and, where available, the use of flow-structure information, and (ii) a broader heat-exchanger toolkit for prediction, optimisation, and diagnostics, where ML serves as a fast surrogate to reduce reliance on repeated experiments and CFD. A recurring conclusion is that progress depends less on identifying a single "best algorithm" and more on consistent datasets, physically meaningful feature sets (often dimensionless), and explicit treatment of applicability limits, uncertainty, and validation protocols.

Yang et al. [1] presented a data-driven perspective focused on rectangular mini-/microchannels. They emphasised that, in confined channels, heat transfer and pressure drop are tightly coupled to evolving flow structures, so purely global regression fits can fail when regimes shift. The review motivates ML workflows that combine careful feature selection and sensitivity analysis with regime-aware data partitioning, while highlighting practical bottlenecks: limited and heterogeneous datasets, inconsistent metadata/reporting, and restricted generalisation across geometries and operating envelopes.

Zou et al. [2] provided a broader heat-exchanger–level view, organising ML usage around performance prediction (surrogate models), design/operation optimisation, and monitoring/diagnostics. Their synthesis stresses limitations that transfer directly to mini-/microchannel boiling and condensation: data scarcity and comparability, sensitivity to extrapolation, and the need for interpretable or physics-guided modelling when the end goal is engineering use rather than in-sample prediction.

Beyond reviews, representative two-phase studies illustrate both the potential and the current constraints of ML-based prediction. Zhou et al. [3] compiled a consolidated multi-source database for flow-condensation HTC spanning multiple fluids and mini-/micro-scale hydraulic diameters, and evaluated ML models for HTC prediction on this heterogeneous dataset. Their results support the methodological value of multi-source aggregation combined with careful validation, while also underscoring that transferability still hinges on strict control of data provenance, feature definitions, and extrapolation limits. Zhu et al. [4] developed an ANN model for HTC prediction in both flow boiling and condensation for minichannels with serrated fins, using a dataset of 1549 experimental points and reporting mean absolute relative deviations of 11.41% (boiling) and 6.06% (condensation). Importantly, they noted that predictive performance is tied to the explored geometry and conditions, which illustrates the practical need for multi-geometry datasets if generalisable models are sought.

Deep-learning approaches have also been explored, including for subcooled flow boiling. Eskandari et al. [5] applied deep-learning models to subcooled boiling in microchannels and reported very high within-dataset accuracy for an LSTM architecture.

Bediako et al. [6] summarised multiple ML families used for boiling HTC prediction (e.g., SVM and ensemble learning) and highlighted a common reporting pattern: strong performance metrics are frequently achieved within a given dataset, whereas clear evidence for cross-geometry and crossregime transfer remains limited, motivating more standardised datasets and experiment-wise validation.

A complementary line of work treats ML as a "generalised correlation" framework based on similarity parameters. Qiu et al. [7] demonstrated this approach for two-phase mini-/microchannel modelling by predicting saturated flow-boiling pressure drop from a consolidated multi-source database. They trained ANN and tree-based ensemble models (XGBoost, LightGBM) on dimensionless inputs and showed that ML can outperform benchmark correlations on random train–test splits, while also revealing a key limitation: generalisation degrades markedly under dataset-level hold-out (especially when a working fluid absent from training is excluded), indicating that correlation-like ML tools still require either overlap in fluids/conditions or explicit mechanisms to encode fluid dependence.

ML models can also support the transition from processed experimental data to compact engineering correlations. In this work, the regressor is not used as a replacement for physics, but as a nonlinear diagnostic tool that helps identify regime-dependent trends in the local Nusselt number and supports the later formulation of compact explicit correlations.

In the present work, local Nusselt number data for flow boiling in rectangular minichannels are analysed using a combined experimental and data-driven approach. The main objective is to derive compact empirical correlations for the subcooled and saturated boiling regimes while avoiding overoptimistic validation caused by dense axial and temporal sampling within individual experiments.

Tree-ensemble regressors are used only as an auxiliary modelling step. Their role is to identify nonlinear dependencies and to support the selection of variables for explicit correlations. The final result is not a black-box ML model, but regime-specific analytical correlations expressed in terms of dimensionless groups and axial position.

1. Experimental data and data reduction

1.1. Experimental facility

The experiments were conducted on a closed-loop flow boiling stand consisting of a dedicated test section, circulation pump, compensating/pressure-control tank, auxiliary heat exchanger, deaeration and filtering components, and a Coriolis mass flow meter. Wall-temperature mapping was performed using infrared thermography. High-speed visualisation of two-phase flow structures through a transparent window, supported by controlled LED lighting and PC-based data acquisition, was used only as qualitative regime context and was not included in the quantitative model inputs.

1.2. Experimental data and methods

Experimental runs obtained on minichannel modules were used to construct local, segment-resolved heat-transfer records. The present paper focuses on the FC-72 smooth-wall series (103 experiments). A smaller FC-770 series (34 experiments) was retained as a reference series and processed according to the same general protocol.

The analysed dataset corresponds to a test module comprising a set of parallel rectangular minichannels. The nominal channel depth is 1 mm and the heated length is approximately 42 mm; the channel width depends on the number of parallel channels in a given module configuration. The channels are asymmetrically heated: one common wall is resistively heated using a thin Haynes-230 foil (0.1 mm thickness) tensioned between structural elements. The outer foil surface is coated with a high-emissivity layer to enable infrared thermography.

During each run, inlet and outlet bulk temperatures were recorded using calibrated thermocouples, while inlet and outlet pressures were measured using pressure transducers; atmospheric pressure was recorded to reconstruct absolute pressure. Electrical current supplied to the foil and the voltage drop across it were measured to compute the supplied power and heat flux based on the heated area. The supplied heat flux was corrected for external losses from the foil surface (natural convection and radiation) using the infrared-measured surface temperature.

Key recorded signals include: (i) the infrared wall-temperature field T_{wall} measured with a FLIR A655SC infrared camera (640 × 480 resolution), (ii) mass flow rate from a Coriolis meter (Endress + Hauser Proline Promass A100), (iii) inlet/outlet gauge pressures from pressure transducers (Endress + Hauser Cerabar S PMP71) together with atmospheric pressure (WIKA A-10) to reconstruct absolute pressure, (iv) inlet/outlet fluid temperatures from calibrated K-type thermocouples, and (v) electrical current and voltage drop across the foil to compute supplied power. Representative absolute uncertainties for this instrumentation are on the order of ±2 (or ±2%) for infrared temperature, ~1.5 K for K-type thermocouples, ±0.05% of reading for gauge pressure, and ±0.1% of reading for mass flow rate.

Data were sampled at 1 s intervals. After degassing and stabilisation, each experiment followed a controlled, manually adjusted increase in electrical power at fixed or quasi-fixed flow conditions, covering single-phase forced convection, boiling incipience, subcooled boiling and saturated boiling.

1.3. Data reduction and regime classification

The data-reduction procedure transformed the raw experimental measurements into local, segment-resolved heat-transfer records. For each experimental run and time frame, the infrared wall-temperature field was reduced to a one-dimensional axial temperature distribution and resampled on a common axial grid. The normalized axial coordinate x was defined so that $x = 0$ corresponds to the inlet and $x = 1$ to the outlet of the observed region. The main processing steps included correction of the supplied heat flux for external heat losses, reconstruction of local pressure and saturation temperature, correction of the effective wall-to-fluid temperature difference for heat conduction through the heater foil, and calculation of the local heat transfer coefficient and Nusselt number.

Local absolute pressure $p_{abs}(x)$ was reconstructed from gauge pressures and atmospheric pressure and interpolated linearly along the channel. The local saturation temperature $T_{sat}(x)$ was computed from $p_{abs}(x)$ using adopted fluid-property correlations for the working fluid. The local bulk-fluid temperature $T_f(x)$ was computed as a linear profile between inlet and outlet temperatures for subcooled operation; for saturated operation, $T_f(x)$ was set to $T_{sat}(x)$.

The local net heat flux was computed from electrical power and heated area and corrected for external heat losses (natural convection and radiation) evaluated from the measured foil surface temperature. A wall-conduction correction through the heater foil was applied to obtain the effective wall-to-fluid temperature difference.

The boiling regime was classified using the wall superheat, $\Delta T = T_{wall} - T_{sat}$. Points with $\Delta T < -2$ K were assigned to the subcooled boiling region, points with $\Delta T > +2$ K to the saturated boiling region, and points within ±2 K were treated as ambiguous and excluded from model fitting. Additional quality-control criteria removed records with non-physical values of heat flux, temperature difference, fluid properties, Reynolds number, Prandtl number or Nusselt number.

Because the complete local dataset contained millions of records, representative quality-controlled subsets of approximately 200,000 points per regime were used for tree-ensemble analysis, whereas final correlation fitting used the larger quality-controlled datasets reported below. Sampling was performed with fixed random seeds to preserve reproducibility.

2. Machine-learning-assisted correlation development

Two tree-ensemble regressors, Extra Trees and Random Forest (scikit-learn), were used to obtain a nonlinear data-driven description of the experimental trends. The models were trained to predict log(Nu), which reduces the influence of large absolute deviations at high Nu values. The input variables were selected to remain compatible with later analytical correlation development and included Reynolds number, Prandtl number, normalized axial position, net heat flux and absolute pressure when available.

Training and validation were performed separately for the subcooled and saturated regimes. For each regime, approximately 200,000 quality-controlled records were sampled for model training and validation, retaining only physically admissible points with positive Nu, Re and Pr. Validation used group splits by experimental run: an entire experiment was assigned either to training or to test to prevent leakage from strongly correlated axial records within the same run. The split was repeated with multiple random seeds to quantify metric variability.

Performance was reported by MAPE on Nu in linear space and by RMSE and R^2 on log(Nu). Feature importances and partial dependence were used qualitatively to guide the selection of an explicit correlation structure; the final publishable result is not the ML model itself but a compact analytical correlation fitted to the experimental data and evaluated using experiment-wise validation.

3. Results and discussion

3.1. Model setup and validation

Model development was performed on the reduced and quality-controlled local heat-transfer dataset, independently of the raw measurement files. The FC-72 dataset contained dense axial and temporal records for each experiment; therefore, random row-wise train–test splitting would have led to information leakage. To avoid this, validation was performed using group splits by experimental run: all rows from a given experiment belonged either to the training set or to the test set.

For the FC-72 smooth-wall series, after quality control and regime labelling, 3,951,814 subcooled and 1,787,425 saturated records were available (originating from 103 and 100 experiments, respectively). Representative subsets of approximately 200,000 points per regime were then sampled while preserving broad coverage of operating conditions. Repeated group-split evaluations with different random seeds were used to quantify the variability of the performance metrics.

The target variable was modelled in log-space, y = log(Nu), which stabilises variance across the wide Nu range. Performance was reported as MAPE on Nu in linear space, and RMSE and R^2 on log(Nu).

3.2. Results for FC-72

Subcooled regime

The analysed subcooled dataset contained 200,026 records from 103 experiments. The subcooled dataset covered Nu ≈ 3.15–90.75, Re ≈ 187–1433, Pr ≈ 11.6–14.0, and net heat flux q_{net} ≈ 0.9–72 kW/m², with absolute pressures in the range ≈ 87–193 kPa. Models were trained on log(Nu) using similarity parameters (Re, Pr), normalized position (x), and variables related to boiling intensity (q_{net}, p_{abs} when available).

Under experiment-wise validation, Extra Trees performed better than Random Forest but the predictive quality remained moderate. Extra Trees gave MAPE = 0.170 ± 0.025, $RMSE_{log}$ = 0.240 ± 0.046 and R^2_{log} = 0.593 ± 0.265, whereas Random Forest gave MAPE = 0.206 ± 0.039, $RMSE_{log}$ = 0.286 ± 0.051 and R^2_{log} = 0.422 ± 0.361. The relatively large variability of R^2_{log} indicates that the subcooled region is sensitive to experiment-to-experiment differences and should not be interpreted as a universally transferable correlation without further validation.

Saturated regime

The analysed saturated dataset contained 200,433 records from 100 experiments. The saturated dataset covered Nu ≈ 3.09–42.95, Re ≈ 313–1519, Pr ≈ 11.6–14.0, and q_{net} ≈ 5.3–105 kW/m².

For the saturated regime, the two ensemble models gave similar MAPE values, although Extra Trees again gave slightly better log-space metrics. Extra Trees achieved MAPE = 0.174 ± 0.050, $RMSE_{log}$ = 0.211 ± 0.050 and R^2_{log} = 0.672 ± 0.098, while Random Forest achieved MAPE = 0.180 ± 0.043, $RMSE_{log}$ = 0.238 ± 0.041 and R^2_{log} = 0.582 ± 0.097. These results show that the ML models capture the main nonlinear trends, but their role in this study is diagnostic and supportive: they provide guidance for the explicit correlation rather than serving as the final predictive tool.

3.3. FC-770 reference series

The FC-770 experiments form a smaller reference series processed according to the same data-reduction and regime-labelling procedure. Because the FC-770 series is smaller than the FC-72 series, it is used here as a reference dataset for applying the same reduction and validation procedure, while the explicit correlations are reported for the larger FC-72 smooth-wall dataset.

4. Proposed correlations, validation and comparison

4.1. Strategy

Tree-ensemble regressors were used as nonlinear diagnostic models supporting the selection of variables in the explicit correlations. The explicit correlations reported here are obtained by fitting compact analytical forms to dimensionless groups computed during data reduction and validating them using experiment-wise group splits.

4.2. Candidate structure and variable selection

Based on feature sweeps, coefficient stability under group-wise validation and physical interpretability for this geometry, regime-specific multiplicative correlations were adopted in terms of Re, Bo and the normalized axial position x, including a quadratic axial term x^2. The Jakob number Ja was included only for the subcooled regime because, in the present formulation based on liquid subcooling, it becomes zero when $T_f = T_{sat}$ in the saturated boiling region. Prandtl number was not retained in the recommended subcooled form because Pr varies weakly in the FC-72 dataset (narrow thermal-property range) and its exponent became unstable or non-beneficial under experiment-wise group validation. Weber number We and explicit pressure terms were excluded from the final recommended forms because the pressure range is relatively narrow and these variables did not yield a stable gain once Re/Bo were included (strong collinearity through definitions).

The following empirical correlations are proposed for the investigated FC-72 smooth-wall dataset, with *x* denoting the normalized axial position:

- for the subcooled boiling regime:

$$\mathrm{Nu} = 2.388\mathrm{Re}^{0.519}\,\mathrm{Bo}^{0.316}\,(1+\mathrm{Ja})^{2.487}\exp(-0.336x+0.212x^2) \tag{1}$$

- for the saturated boiling regime:

$$\mathrm{Nu} = 5.285\mathrm{Re}^{0.885}\,\mathrm{Pr}^{0.490}\mathrm{Bo}^{0.866}\exp(0.108x+0.073x^2) \tag{2}$$

4.3. Fitting and validation

Coefficients and exponents were obtained by log-linear regression in log(Nu), using repeated group splits by experimental run to prevent leakage from dense axial sampling within a single experiment. For the final correlation fits, the training tables contained 870,198 (subcooled, 100 experiments) and 839,820 (saturated, 95 experiments) records.

For the subcooled regime, the 10-split group cross-validation (group-CV) performance was MAPE = 0.184982 ± 0.038334, $RMSE_{log}$ = 0.230951 ± 0.038591 and R^2_{log} = 0.605212 ± 0.116036. For the saturated regime, the corresponding values were MAPE = 0.074055 ± 0.006552, $RMSE_{log}$ = 0.097649 ± 0.010304 and R^2_{log} = 0.909833 ± 0.037183. Direct evaluation of the final saturated-regime correlation on the complete saturated-regime fitting dataset gave MAPE = 0.06940, $RMSE_{log}$ = 0.08971 and R^2_{log} = 0.94218, confirming that this was the most stable correlation obtained in the present study.

For the subcooled regime, the lower R^2_{log} and larger split-to-split variability indicate that the proposed correlation should be used only within the investigated range. Its extrapolation towards single-phase forced convection or towards other channel geometries requires separate validation.

Conclusions

This study developed an ML-assisted procedure for deriving empirical Nusselt number correlations from local flow-boiling data obtained in rectangular minichannels. The main contribution is methodological and correlation-oriented: the study combines local experimental data, experiment-wise validation and explicit regime-specific Nu correlations. The analysis focused on the FC-72 smooth-wall experimental series, while FC-770 was retained as a smaller reference series processed using the same general protocol. In the present paper, the explicit fitted correlations are reported for the larger FC-72 dataset, whereas the FC-770 series is retained as a reference case for applying the same processing and validation framework to a second working fluid.

Experiment-wise validation showed that tree-ensemble models can reproduce the main nonlinear trends in the local Nu data, but the obtained metrics also reveal substantial experiment-to-experiment variability. Therefore, the ML models were not treated as final engineering correlations, but as diagnostic tools supporting the selection of compact analytical forms.

For the FC-72 dataset, the saturated-boiling correlation gave the most stable result and the lowest prediction error, with 10-split group-validation performance of MAPE = 0.074 ± 0.007 and R^2_{log} = 0.910 ± 0.037. The subcooled-boiling correlation was less stable, with MAPE = 0.185 ± 0.038 and R^2_{log} = 0.605

± 0.116, and should therefore be regarded as an empirical approximation restricted to the investigated operating range.

The proposed correlations are applicable to the present rectangular minichannel geometry, smooth heated wall and investigated FC-72 operating conditions. Further validation is required before transferring the correlations to other fluids, enhanced surfaces, channel geometries or wider pressure and heat-flux ranges.

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Nomenclature

α	heat transfer coefficient (W m^{-2} K^{-1})
ΔT	wall superheat, $T_{wall} - T_{sat}$ (K)
Bo	boiling number (-)
Ja	Jakob number (-)
MAPE	mean absolute percentage error (%)
Nu	Nusselt number (-)
p_{abs}	absolute pressure (Pa)
Pr	Prandtl number (-)
q_{net}	net heat flux (W m^{-2})
R^2_{log}	coefficient of determination in log-space (-)
Re	Reynolds number (-)
$RMSE_{log}$	root-mean-square error in log-space (-)
T_f	bulk-fluid temperature (K or °C, according to the processing stage)
T_{sat}	saturation temperature (K or °C, according to the processing stage)
T_{wall}	wall temperature (K or °C, according to the processing stage)
We	Weber number (-)
x	normalized axial coordinate (-)

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